



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2008

YEAR 11
Mathematics (2U)
Half Yearly Examination

Mathematics

General Instructions

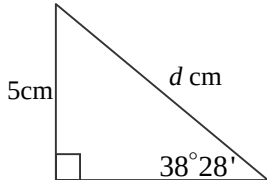
- Write using black or blue pen
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.

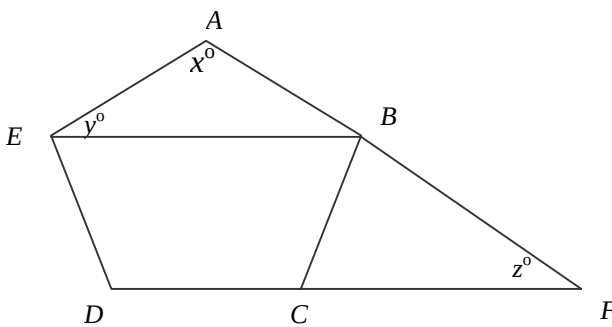
Reading time: 5 minutes
Working time: 90 minutes

Total Marks – 70

- Attempt All Questions.
- All questions are NOT of equal value.

Examiner: *A. Ward*

Number	Question	Marks
1.	Find 0.009738148 correct to three significant figures.	1
2.	Write down the supplement of $126^{\circ}52'$.	1
3.	Simplify $-5p^4 + 3p^2 + 2p^4$.	1
4.	Simplify $2\frac{1}{2} \times 3\frac{7}{8} \div 5\frac{5}{9}$.	1
5.	Simplify $\sqrt{27} + \sqrt{48}$.	1
6.	Expand and simplify $(2x-3)(3x+4)$.	1
7.	Solve $\frac{6t-1}{t} = 4$.	1
8.	Factorise $x^2 - 13x + 30$.	1
9.	Calculate: $\frac{\sqrt{4.7-3.2}}{9.4-1.6}$ correct to 3 decimal places.	1
10.	Express $0.\dot{2}\dot{1}$ as a rational number.	1
11.	Solve $x^2 - 4x = 0$.	1
12.	Find the value of d to 3 decimal places.	
		1
13.	Write down the exact value of $\cos 210^{\circ}$.	1
14.	Solve $(x-4)^2 = 17$. Leave your answer in surd form.	1
15.	Express $\frac{7\pi}{15}$ radians in degrees.	1
16.	Simplify $\frac{5x}{x^2-4} - \frac{x+2}{10x^2}$.	2
17.	Express in $\sqrt{1452}$ the form $a\sqrt{b}$ where $\sqrt{b}$ is irrational.	2
18.	Find c in terms of a , b and d :	
	$d = \sqrt{b^2 - 4ac}$	2

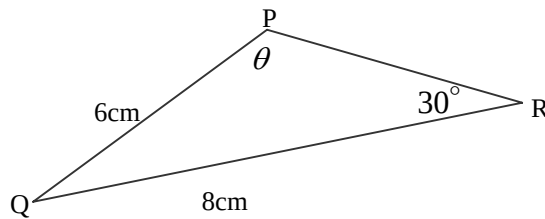
19. Factorise fully:
- $rs - 15 + 3s - 5r$ 2
 - $2x^2 + x - 21$ 2
 - $x^4 - a^4$ 2
 - $x^3 - a^3$ 2
20. Solve $2x^2 + 8x + 16 = 0$ by completing the square. 2
21. If $\cos \theta = \frac{-7}{25}$ and $90^\circ \leq \theta \leq 180^\circ$, find $\tan \theta$. 2
22. Solve $3(2 + y) - 4(2y - 7) \leq 9$. 2
23. Express with a rational denominator:
- $$\frac{2\sqrt{5} + 1}{2\sqrt{5} - 1}$$
- 2
24. Solve: $|2x - 5| = 7$ 2
25. Simplify $(x + a)^2 + (x - a)^2$ 2
26. Solve $x + \frac{2}{x} = -3$. 2
27. Solve the simultaneous equations:
- $$\begin{aligned} x + y + z &= 7 \\ 2x + 3y - z &= 0 \\ 3x + 4y + 2z &= 17 \end{aligned}$$
- 3
28. A triangle PQR whose area is 3cm^2 has sides $PQ = 4\text{ cm}$ and $QR = \sqrt{3}\text{ cm}$. Find $\angle PQR$. 3
29. $ABCDE$ is a regular pentagon, with AB and DC produced to F . Find the values of x , y and z . (Do not give reasons)
- 
- 3
30. In triangle RPT , $r = 15$, $t = 8$ and $\angle RPT = 2$ radians. Find side p to 2 decimal places. 3

31. Solve the inequality:

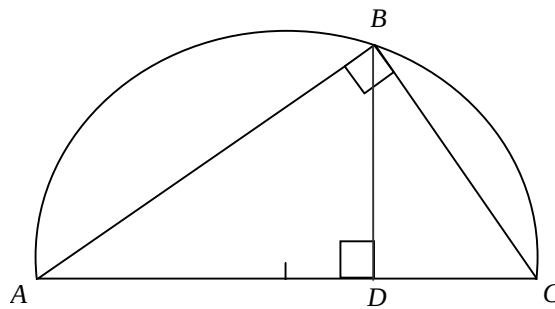
$$|x + 3| \leq 2x - 2 \quad 3$$

32. If $p = \frac{\cos \theta}{2}$ and $q = \sin^3 \theta$, find $\sin^2 \theta + \cos^2 \theta$ in terms of p and q . 3

33. In triangle PQR , the obtuse angle $\angle QPR = \theta$. The angle $\angle PRQ = 30^\circ$ and the lengths PQ and QR are 6cm and 8cm respectively. Find the value of $\cos \theta$ correct to 3 significant figures.



- 34.



In the diagram AC is the diameter of a circle radius r and BD is perpendicular to AC .

- (a) Prove that triangles ABC , ADB and BDC are similar. 2
- (b) Given that $BC = \frac{1}{2}r$ show that 3
 $AD:DC = 15:1$

END OF EXAMINATION

Year 11 2 unit $\frac{1}{2}$ yearly 2008

1. 0.009738148 3 s.f

$$= 0.00974 \quad (1)$$

2. Supp of $126^\circ 52'$
 $= 180^\circ - 126^\circ 52'$
 $= 53^\circ 8' \quad (1)$

3. $-5p^4 + 3p^2 + 2p^4$
 $= 3p^2 - 3p^4 \quad (1)$

4. $2\frac{1}{2} \times 3\frac{7}{8} \div 5\frac{5}{9}$
 $= 1\frac{19}{160} \text{ OR } 279/160 \quad (1)$

5. $\sqrt{27} + \sqrt{48}$
 $= \sqrt{9} \cdot \sqrt{3} + \sqrt{16} \cdot \sqrt{3}$
 $= 3\sqrt{3} + 4\sqrt{3}$
 $= 7\sqrt{3} \quad (1)$

6. $(2x - 3)(3x + 4)$
 $= 6x^2 + 8x - 9x - 12$
 $= 6x^2 - x - 12 \quad (1)$

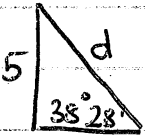
7. $\frac{6t-1}{t} = 4$
 $6t - 1 = 4t$
 $2t = 1$
 $t = \frac{1}{2} \quad (1)$

8. $x^2 - 13x + 30$
 $= (x - 3)(x - 10) \quad (1)$

9. $\frac{\sqrt{457-302}}{9-4-16}$
 $= 0.157018573$
 $= 0.157 \text{ 3dp} \quad (1)$

10. $x = 0.\dot{2}\dot{1}$
 $100x = 21.\dot{2}\dot{1}$
 $99x = 21$
 $x = \frac{21}{99} = \frac{7}{33} \quad (1)$

11. $x^2 - 4x = 0$
 $x(x - 4) = 0$
 $x = 0, x - 4 = 0$
 $(\frac{1}{2}) \quad x = 4 \quad (\frac{1}{2})$

12.  $\frac{d}{\sin 90} = \frac{5}{\sin 38^\circ 28'}$

$$d = 8.037819828$$
$$= 8.038 \text{ 3dp} \quad (1)$$

13. $\cos 210^\circ = -\frac{\sqrt{3}}{2} \quad (1)$

14. $(x - 4)^2 = 17$
 $x - 4 = \pm \sqrt{17}$
 $x = 4 \pm \sqrt{17} \quad (1)$

15. $\frac{7\pi}{15}$ in degrees
 $\frac{7\pi}{15} \times \frac{180}{\pi} = 84^\circ \quad (1)$

$$\begin{aligned}
 16. \quad & \frac{5x}{x^2-4} - \frac{x+2}{10x^2} \\
 = & \frac{5x}{(x+2)(x-2)} - \frac{x+2}{10x^2} \\
 = & \frac{50x^3 - (x+2)^2(x-2)}{10x^2(x+2)(x-2)} \quad ① \\
 = & \frac{50x^3 - (x^2+4x+4)(x-2)}{10x^2(x+2)(x-2)} \\
 = & \frac{50x^3 - (x^3-2x^2+4x^2-8x+4x-8)}{10x^2(x+2)(x-2)} \\
 = & \frac{50x^3 - x^3 - 2x^2 + 4x + 8}{10x^2(x+2)(x-2)} \\
 = & \frac{49x^3 - 2x^2 + 4x + 8}{10x^2(x+2)(x-2)} \quad ①
 \end{aligned}$$

$$\begin{aligned}
 17. \quad & \sqrt{1452} = a\sqrt{b} \\
 & \sqrt{4} \cdot \sqrt{121} \cdot \sqrt{3} = a\sqrt{b} \\
 & 2 \cdot 11 \cdot \sqrt{3} = a\sqrt{b} \\
 & 22\sqrt{3} = a\sqrt{b} \\
 \therefore a = 22 \quad b = 3 \\
 & \quad \quad \quad ① \quad \quad \quad ①
 \end{aligned}$$

$$\begin{aligned}
 18. \quad & d = \sqrt{b^2 - 4ac} \\
 & d^2 = b^2 - 4ac \quad ① \\
 & 4ac = b^2 - d^2 \\
 & c = \frac{b^2 - d^2}{4a} \quad ①
 \end{aligned}$$

$$\begin{aligned}
 19. a) \quad & rs - 15 + 3s - 5r \\
 & s(r+3) - 5(r+3) \quad ① \\
 & = (r+3)(s-5) \quad ①
 \end{aligned}$$

$$\begin{aligned}
 19. b) \quad & 2x^2 + x - 21 \\
 & \frac{(2x+7)(2x-6)}{2} \\
 & = \frac{2(x-3)(2x+7)}{2} \\
 & = (x-3)(2x+7) \quad ②
 \end{aligned}$$

$$\begin{aligned}
 c) \quad & x^4 - a^4 \\
 & = (x^2 - a^2)(x^2 + a^2) \quad ① \\
 & = (x^2 + a^2)(x+a)(x-a) \quad ①
 \end{aligned}$$

$$\begin{aligned}
 d) \quad & x^3 - a^3 \\
 & = (x-a)(x^2 + xa + a^2) \quad ②
 \end{aligned}$$

$$\begin{aligned}
 20. \quad & 2x^2 + 8x + 16 = 0 \\
 & x^2 + 4x + 8 = 0
 \end{aligned}$$

$$\text{~~(x+2)^2 = -4~~}$$

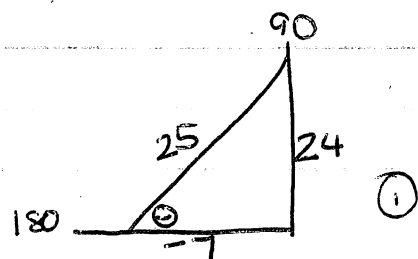
$$x^2 + 4x + 4 = -8 + 4$$

$$(x+2)^2 = -4$$

$$x+2 = \sqrt{-4}$$

$$\therefore \text{no solutions} \quad ②$$

$$21. \cos \theta = -7/25$$



$$\tan \theta = -\frac{24}{7} \quad ①$$

$$\begin{aligned}
 22. \quad & 3(2+y) - 4(2y-7) \leq 9 \\
 & 6 + 3y - 8y + 28 \leq 9 \quad \textcircled{1} \\
 & 34 - 5y \leq 9 \\
 & -5y \leq -25 \\
 & y \geq 5 \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 23. \quad & \frac{2\sqrt{5}+1}{2\sqrt{5}-1} \times \frac{2\sqrt{5}+1}{2\sqrt{5}+1} \quad \textcircled{1} \\
 & = \frac{20 + 4\sqrt{5} + 1}{20 - 1} \\
 & = \frac{21 + 4\sqrt{5}}{19} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 24. \quad & |2x - 5| = 7 \\
 & 2x - 5 = 7 \quad 2x - 5 = -7 \\
 & 2x = 12 \quad 2x = -2 \\
 & x = 6 \quad x = -1 \\
 & \textcircled{1} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 25. \quad & (x+a)^2 + (x-a)^2 \\
 & x^2 + 2xa + a^2 + x^2 - 2xa + a^2 \quad \textcircled{1} \\
 & = 2x^2 + 2a^2 \quad \textcircled{1} \\
 & = 2(x^2 + a^2) \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 26. \quad & x + \frac{2}{x} = -3 \\
 & x^2 + 2 = -3x \\
 & x^2 + 3x + 2 = 0 \\
 & (x+2)(x+1) = 0 \\
 & x = -2, -1 \\
 & \textcircled{1} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 27. \quad & x + y + 3 = 7 \quad \textcircled{1} \\
 & 2x + 3y - 3 = 0 \quad \textcircled{2} \\
 & 3x + 4y + 23 = 17 \quad \textcircled{3}
 \end{aligned}$$

$$\begin{aligned}
 & \textcircled{2} \times 2 \\
 & 4x + 6y - 23 = 0 \quad \textcircled{4}
 \end{aligned}$$

$$\begin{aligned}
 & \textcircled{4} + \textcircled{3} \quad 7x + 10y = 17 \quad \textcircled{5} \\
 & \textcircled{1} + \textcircled{2} \quad 3x + 4y = 7 \quad \textcircled{6}
 \end{aligned}$$

$$\begin{aligned}
 & \textcircled{5} \times 3 \quad 21x + 30y = 51 \quad \textcircled{7} \\
 & \textcircled{6} \times 7 \quad 21x + 28y = 49 \quad \textcircled{8} \\
 & \textcircled{7} - \textcircled{8}
 \end{aligned}$$

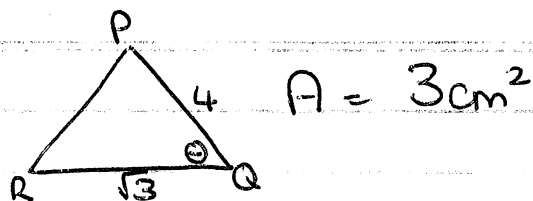
$$\begin{aligned}
 & 2y = 2 \\
 & \boxed{y = 1} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 & \text{into } \textcircled{6} \quad 3x + 4 = 7 \\
 & 3x = 3 \\
 & \boxed{x = 1} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 & \text{into } \textcircled{1} \\
 & 1 + 1 + 3 = 7 \\
 & \boxed{3 = 5} \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 & \text{test in } \textcircled{2} \\
 & 2 + 3 - 5 = 0 \quad \checkmark
 \end{aligned}$$

28.



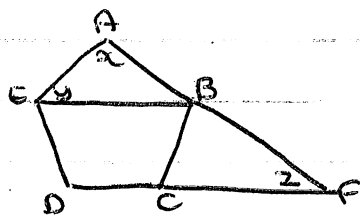
$$A = \frac{1}{2}ab \sin C \quad (1)$$

$$3 = \frac{1}{2} \times 4 \times \sqrt{3} \times \sin \theta$$

$$\sin \theta = \frac{6}{4\sqrt{3}} \quad (1)$$

$$\theta = 60^\circ \quad (1)$$

29.



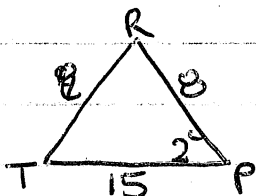
$$\text{regular pentagon} = 540^\circ$$

$$\therefore x = 108^\circ \quad (1)$$

$$y = 36^\circ \quad (1)$$

$$z = 36^\circ \quad (1)$$

30.



$$p^2 = 8^2 + 15^2 - 2 \times 8 \times 15 \times \cos 2^\circ$$

$$= 289 - 240 \cos 2^\circ$$

$$p = \sqrt{289 - 240 \cos 2^\circ}$$

$$= 13.75226379$$

$$= 13.75 \quad (3)$$

deg 7.01

31. $|x+3| \leq 2x-2$

$$x+3 \leq 2x-2 \quad -x-3 \geq 2x-2$$

$$5 \leq x$$

$$x \geq 5$$

(1/2)

$$-1 \geq 3x$$

$$x \leq -1/3$$

(1/2)

32.

$$p = \frac{\cos \theta}{2}, \quad q = \sin^3 \theta$$

$$2p = \cos \theta$$

$$4p^2 = \cos^2 \theta$$

(1)

$$q = \sin^3 \theta$$

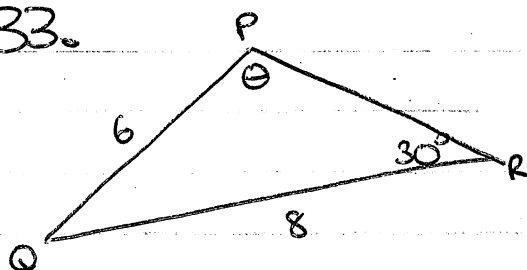
$$\sin^2 \theta = \frac{q}{\sin \theta}$$

(1)

$$\cos^2 \theta + \sin^2 \theta = 4p^2 + \frac{q^2}{\sin^2 \theta} \quad (1)$$

(1/2)

33.



$$\frac{\sin 30}{6} = \frac{\sin \theta}{8} \quad (1)$$

$$\sin \theta = \frac{8 \sin 30}{6}$$

$$\sin \theta = \frac{2}{3}$$

$$\sin \theta = \frac{2}{3}$$

(1/2)

$$\theta = 41.8103149^\circ \quad (2)$$

$$\cos \theta = 0.745355992$$

$$= 0.745 \text{ (3 s.f.)} \quad (1/2)$$

34. a) let $\angle A = x$.

$$\triangle ABC \text{ and } \triangle ABD$$

$$\angle ABC = 90^\circ = \angle ADB$$

$$\angle A \text{ common}$$

$$\therefore \angle ABB = 90 - \angle A = \angle ABD$$

$$\therefore \triangle ABC \text{ and } \triangle ABD \text{ are similar}$$

$$\triangle ABD \sim \triangle BDC$$

$$\angle ADB = 90^\circ = \angle BDC$$

$$\angle C = 90^\circ - \angle A \text{ (from above)}$$

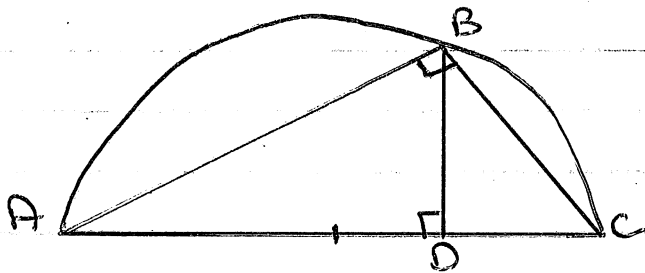
$$\therefore \angle ABD = 90^\circ - \angle A$$

$$\therefore \angle BAD = \angle CBD$$

$$\therefore \triangle ABD \sim \triangle BDC$$

$$\therefore \triangle ABC \sim \triangle ADB \sim \triangle BDC \quad (2)$$

b)



$$\text{radius} = r \text{ (given)}$$

$$BC = \frac{1}{2}r$$

$$AC = 2r \text{ (diameter)}$$

$$\therefore (2r)^2 = \left(\frac{r}{2}\right)^2 + AB^2$$

$$AB^2 = 4r^2 - \frac{r^2}{4}$$

$$= \frac{15}{4}r^2$$

$$AB = \frac{\sqrt{15}}{2}r$$

$$AC : AB = AB : AD$$

(similar Δ 's sides same ratio)

$$2r : \frac{\sqrt{15}}{2}r = \frac{\sqrt{15}}{2}r : AD$$

$$2 : \frac{\sqrt{15}}{2} = \frac{\sqrt{15}}{2}r : AD$$

$$1 : \frac{\sqrt{15}}{4} = \frac{\sqrt{15}}{2}r : AD$$

$$\therefore AD = \frac{\sqrt{15}}{4}r \cdot \frac{\sqrt{15}}{2}r$$

$$AD = \frac{15}{8}r$$

In $\triangle ABD$

$$AB^2 = AD^2 + BD^2$$

$$\frac{15}{4}r^2 = \frac{225}{64}r^2 + BD^2$$

$$BD^2 = \frac{15}{64}r^2$$

$$BD = \frac{\sqrt{15}}{8}r$$

In $\triangle BCD$

$$BC^2 = BD^2 + DC^2$$

$$\frac{r^2}{4} = \frac{15}{64}r^2 + DC^2$$

$$DC^2 = \frac{1}{64}r^2$$

$$DC = \frac{1}{8}r$$

$$\therefore AD : DC$$

$$= \frac{15r}{8} : \frac{1r}{8}$$

$$= 15 : 1$$

(3)

(I'm sure there is an easier way to do this!)